

Quantum Descriptors Reference

Feature	Formula / definition	Description
energy_hartree	E0 (Hartree)	Total GFN2-xTB electronic energy for the geometry (use relative energies within a molecule).
homo_eV	EHOMO (eV)	HOMO energy; electron-donor strength proxy.
lumo_eV	ELUMO (eV)	LUMO energy; electron-acceptor strength proxy.
gap_eV	ELUMO - EHOMO	Frontier-orbital gap; crude proxy for absorption onset.
mu_eV	$\mu = (EHOMO + ELUMO) / 2$	Chemical potential (conceptual DFT).
eta_eV	$\eta = (ELUMO - EHOMO) / 2$	Chemical hardness (conceptual DFT).
softness_1_per_eV	$S = 1 / \eta$	Softness; inverse hardness (proxy for electronic polarizability).
chi_eV	$\chi = -\mu$	Electronegativity (conceptual DFT).
omega_eV	$\omega = \mu^2 / (2\eta)$	Electrophilicity index (charge-transfer tendency).
dipole_D	$dipole = \sqrt{dx^2 + dy^2 + dz^2} * 2.541746$	Dipole moment magnitude (Debye).
quad_norm_au	$quad_norm = \sqrt{\sum(Q_i^2)}$	Quadrupole tensor norm (charge anisotropy).
quad_trace_au	$quad_trace = Q_{xx} + Q_{yy} + Q_{zz}$	Quadrupole trace (overall 2nd-order charge spread).
bo_sum	$bo_sum = \sum_{i<j}(BO_{ij})$	Sum of bond orders over all atom pairs (upper triangle).
bo_max	$bo_max = \max_{i<j}(BO_{ij})$	Maximum bond order over all atom pairs.
bo_mean_bonds	mean BO over RDKit bonded pairs	Mean BO restricted to pairs connected by an RDKit bond.
bo_conj_sum	sum BO over conjugated/aromatic RDKit bonds	Sum of BO over bonds marked conjugated/aromatic (proxy for pi-delocalization).
bo_conj_mean	mean BO over conjugated/aromatic RDKit bonds	Mean BO over conjugated/aromatic bonds.
q_min	$\min_i(q_i)$	Minimum atomic partial charge.
q_max	$\max_i(q_i)$	Maximum atomic partial charge.
q_mean	$\text{mean}_i(q_i)$	Mean atomic partial charge.
q_std	$\text{std}_i(q_i)$	Std of atomic partial charges.
q_abs_sum	$\sum_i(q_i)$	Total absolute charge (polarization measure).
q_range	$q_{max} - q_{min}$	Charge separation amplitude.
q_pos_top3_mean	mean(top 3 of q_i where $q_i > 0$)	Mean of three largest positive charges.
q_neg_top3_mean	mean(top 3 most negative q_i where $q_i < 0$)	Mean of three most negative charges.
q_abs_r_mean	$\sum_i(q_i * r_i) / \sum_i(q_i)$	Charge-weighted mean radius (r_i from centroid).
q_abs_r2_rms	$\sqrt{\sum_i(q_i * r_i^2) / \sum_i(q_i)}$	Charge-weighted RMS radius.
d_pos_neg	$ r_{plus} - r_{minus} $	Distance between charge-weighted centroids of positive vs negative charge.
vip_eV	$VIP = (E_{plus} - E_0) * 27.211386$	Vertical ionization potential via Delta-SCF ($q=+1$).

vea_eV	$VEA = (E_0 - E_{\text{minus}}) * 27.211386$	Vertical electron affinity via Delta-SCF ($q=-1$).
fplus_max	$\max_i(q_i(N+1) - q_i(N))$	Max Fukui $f+$ (nucleophilic reactivity peak).
fplus_sum_pos	$\sum_i(fplus_i \text{ for } fplus_i > 0)$	Total positive $f+$ (overall nucleophilic tendency).
fplus_top3_mean	$\text{mean}(\text{top 3 of } fplus_i \text{ where } fplus_i > 0)$	Dominant nucleophilic sites (top-3 mean).
fminus_max	$\max_i(q_i(N) - q_i(N-1))$	Max Fukui $f-$ (electrophilic reactivity peak).
fminus_sum_pos	$\sum_i(fminus_i \text{ for } fminus_i > 0)$	Total positive $f-$ (overall electrophilic tendency).
fminus_top3_mean	$\text{mean}(\text{top 3 of } fminus_i \text{ where } fminus_i > 0)$	Dominant electrophilic sites (top-3 mean).